Day 3

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Underwater Image Enhancement Completed Work

Work flow

- Raw underwater image dataset collected and inspected.
- Dataset quality and image characteristics analyzed.
- Input–target image pairing process developed.
- Incorrect/mismatched data identified during visual checking.
- Dataset cleaning and Dataset V1 preparation performed.
- Classical enhancement methods implemented:
  - CLAHE
  - Gamma Correction
  - White Balance
- PSNR, SSIM and Edge F1 metrics calculated.
- Learning-based CNN enhancement model trained.
- Model refinement and error/failure analysis performed.

Pipeline

Raw Dataset
↓
Quality check
↓
Pairing
↓
Preprocessing
↓
Training
↓
Enhancement
↓
Evaluation
↓
Error analysis

Findings & Blockers

Key Findings

- Underwater input images generally have lower brightness and contrast than target images.
- Classical enhancement methods provide useful baseline results.
- CNN-based enhancement produced higher PSNR and SSIM than the initial classical baselines.
- Edge preservation remains an important evaluation aspect.
- Different methods show different behavior depending on the underwater scene.
- Visual inspection is necessary because numerical metrics alone do not fully describe enhancement quality.

Blockers / Challenges

- Some input-target pairs were found to be incorrectly matched.
- Dataset quality can directly affect model performance.
- Severely dark and low-contrast scenes are difficult to enhance.
- There can be a trade-off between image similarity and edge preservation.
- Limited correctly paired data may affect generalization

Next Steps

Evaluation Metrics

- PSNR
- SSIM
- Edge F1 / Edge Preservation
- Processing Time
- Visual comparison

Next Plan

- Finalize the verified dataset and frozen pipeline.
- Complete controlled model/parameter experiments.
- Compare final candidate with classical baseline.
- Perform robustness and ablation analysis.
- Complete final validation and failure analysis.
- Prepare final report and presentation.

Thank You